University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering Part-I Odd Semester Examination 2020
Course: EEE-1131 (Basic Electronics) (2019-2020)
Time: 03 Hours Full Marks: 52.5

[N.B. Answer 06 (Six) questions taking any 03 (Three) questions from each section. The figures in the right margin indicate full marks.]

Section-A

1. (a) Define conductor, semiconductor and insulator in terms of energy band diagram. 3
   (b) Discuss the formation process of P-type and N-type extrinsic semiconductor with proper diagram. 2.75
   (c) Consider a specimen of Silicon of length 1.5 cm and area 1 mm^2. Calculate the i) conductivity ii) resistivity and iii) resistance of the specimen. (Assume n_i=1.5 x10^{16}, μ_e = 0.13 m^2/v-s, μ_h = 0.05 m^2/v-s) 3

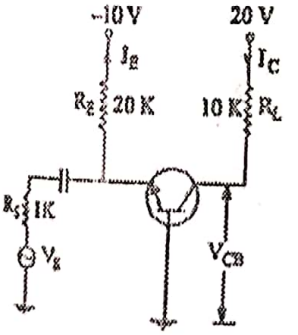
2. (a) What is a p-n junction diode? How can you use a junction diode as a rectifier? Explain. 3
   (b) Draw the V-I characteristic of a p-n junction diode. 3
   (c) Discuss the formation of depletion layer in a PN junction diode. 2.75

3. (a) Differentiate between Zener breakdown and avalanche breakdown. 2
   (b) Show the V-I characteristics of a Zener diode. 2.75
   (c) Zener diode can be used as a voltage regulator. Explain. 4

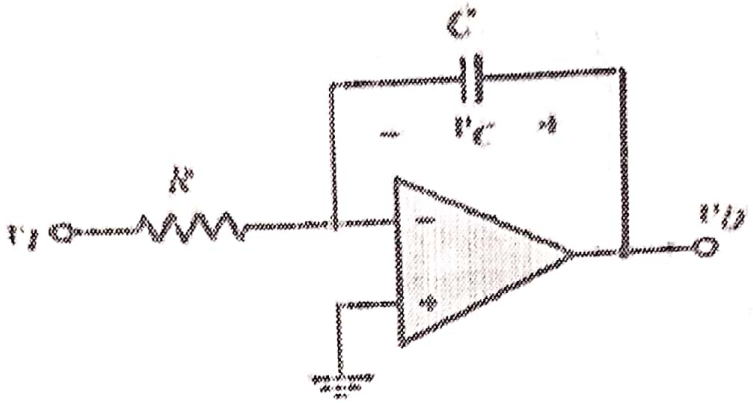
4. (a) What is a transistor? Discuss the structure of a transistor. 3
   (b) How a transistor can be used as switch? Explain. 3
   (c) What do you mean by transistor biasing? Why it is required? 2.75

Section-B

5. (a) Show the relationship between α and β (in case of BJT). 1.75
   (b) Draw the output characteristic of a CE amplifier and show the cut off, active and saturation region. 3
   (c) Draw the dc load line and find the Q point of the following circuit. 4



6. (a) What is an OP-AMP? What are the basic characteristics of an ideal OP-AMP? 3
(b) What do you mean by virtual ground of an OP-AMP? Explain. 1.75
(c) A 10mV, 5KHz sinusoidal signal is applied to the input of an OP-AMP integrator as shown below for which R= 100K and C= 1μF. Find the output voltage. 4



7. (a) What is FET and what are its types? 1.75
(b) Describe the basic operation of a MOSFET. Define enhancement mode and depletion mode. 3
(c) Describe the general current-voltage characteristics for both enhancement mode and depletion mode MOSFETs. 4

8. (a) What is a photodiode? Explain different types of photodiodes according to their function and structure? 3.75
(b) Explain the V-I characteristics of a photodiode. 3
(c) What are the applications of phototransistors? 2

University of Rajshahi
(Affiliated College)
Department of Computer Science and Engineering
B.Sc. Engg. 1st Year Odd Semester Examination- 2021
Course: EEE 1131 (Basic Electronics)
Time: 3 Hrs. Full Marks: 60

[N.B. Answer any *three* questions from each section.]

Section A

1. a) What is semiconductor? Discuss how temperature and doping affect the conductivity of a pure semiconductor. 4
b) What is recombination? Explain whether or not a P-type semiconductor is positively charged. 3
c) A PN junction diode has a leakage current of 10^{-15} A at 25° C. If the diode is forward biased and current through the diode is 5 mA at that temperature then calculate the value of barrier potential. 3

2. a) What is a rectifier? Show the circuit diagram of a full wave rectifier. 4
b) Define voltage regulation (VR) and Peak Inverse Voltage (PIV). 2
c) A half wave rectifier using Ge diode has secondary emf of 20 V_{rms}. Diode forward resistance is 0.25 Ohm and load resistance is 1000 Ohm. Find i) Maximum load voltage ii) DC load voltage and iii) efficiency 4

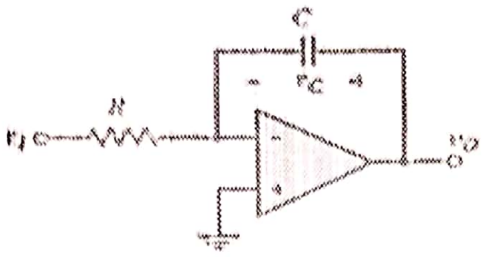
3. a) Explain the working principle of a PNP bipolar junction transistor (BJT) with proper diagram. 3
b) Draw the DC input and output characteristics of an NPN BJT in common emitter configuration. Also show the different regions in the output characteristics curve. 5
c) How transistor can be used as a switch? 2

4. a) Why transistor biasing is necessary? Discuss any one method used for transistor biasing. 5
b) For a certain transistor, I_C= 5.505 mA, I_B=50 μA, I_{CO}= 5 μA. Determine 3
i) the value of α, β and I_E.
ii) the new level of I_B required to make I_C = 10 mA.
c) Show the relationship between α and β for a transistor. 2

Section B

5. a) What is an oscillator? State the conditions for oscillation? 3
b) Explain feedback. 2
c) Design an Astable Multivibrator whose frequency of oscillation is 20 KHz. Consider $R_1 = R_2$ and $C_1 = C_2$. 5

6. a) What is an Op Amp? What are the basic characteristics of an ideal OP-AMP? 4
b) What do you mean by virtual ground of an op-amp? Explain. 2
c) A 10mV, 1KHz sinusoidal signal is applied to the input of an OP-AMP integrator as shown below for which R= 10K and C= 1μF. Find the output voltage. 4



7. a) Classify transistor in terms of charge carrier. Which of the components you listed are current controlled device and which are voltage controlled device? 3
b) With proper diagram discuss the basic architecture of a JFET and it's working principle. 4
c) What is pinch-off voltage? Differentiate between active and passive components. 3

8. a) What is LCD? Describe the working principle of LCD. 5
b) Compare LCD and LED? 2
c) What is P-N photo-diode? Why it works in reverse biased condition? 3

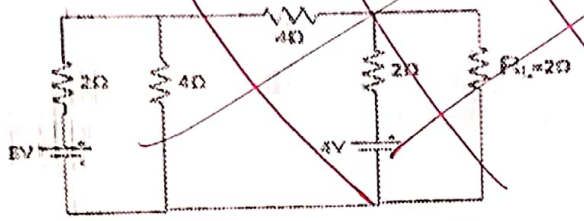
University of Rajshahi
Department of Computer Science and Engineering
B. Sc. Engg. Part-1 Odd Semester, Examination-2021
Course: EEE 1131 (Electrical Circuit and Electronics), Session 2018-2019
APEE 1131 (Electrical Circuit and Electronics), Session 2017-2018

Full Marks: 52.5 Duration: 3 Hours

[Answer Six questions taking at least Three from each section]

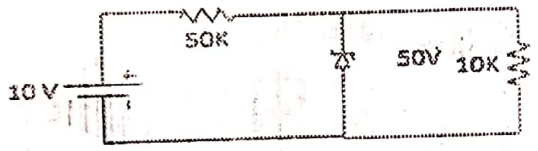
Section A

1. (a) State and explain Thevenin's theorem. 5
(b) Using Thevenin's theorem, find the current through the R_L=2Ω as in the following figure: 3.75



2. (a) What do you mean by doping and dopant? 3
(b) Explain the life time of a carrier. 2
(c) What length of a round copper wire of diameter 1 mm will have a resistance of 1 kΩ if the conductivity of copper is 60 MS/m? A cylindrical piece of silicon having a diameter of 1 mm is doped with 10^{20} /m3 atoms of phosphorous. What length of this silicon would be required to give a resistance of 1 kΩ if the electron mobility in silicon is 0.1 m^2 /V-s? 3.75

3. (a) What are the differences between a Zener diode and a PN junction diode? 2
(b) Explain the V-I characteristics of a Zener diode. 2.75
(c) For the circuit shown below, find: 4



i) Output voltage
ii) Voltage drop across 50 kΩ resistor.
iii) Voltage across the diode.

4. (a) Draw the circuit diagram of an NPN transistor in CE configuration and discuss its input, output and constant current static characteristics. 3.75
(b) What is load line? Show the importance of load line with proper diagram. 2
(c) What is leakage current? 1
(d) What is thermal Runaway? Define stability factor. 2

Section B

5.	(a) Compare between damped and undamped oscillation.	2.75
	(b) What is a multivibrator? Classify multivibrators.	2
	(c) Design an Astable Multivibrator whose frequency of oscillation is 5 KHz. Consider $R_1 = R_2$ and $C_1 = C_2$.	4
6.	(a) What is an operational Amplifier? What do you know about input bias current, input offset voltage and input offset current?	3
	(b) What are CMRR and slew rate? Write some importance of these terms.	2
	(c) Draw the basic circuit of an inverting amplifier and summing amplifier and find out their output voltage and gain.	3.75
7.	(a) What is an oscillator? What are different types of oscillator?	2
	(b) Explain the conditions for oscillation.	2
	(c) Design a Hartley oscillator which has to be tunable over 500 kHz to 1000 kHz. The values of the two inductors are 100 μH each. Neglect the effect of mutual inductance.	4.75
8.	(a) What is LED? Describe the working principle of LED.	4.75
	(b) Why Si or Ge is not used for LED?	2
	(c) What is P-N photodiode? Why it works in reverse biased condition?	2

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engineering Part-I Odd Semester Examination 2020

Course: EEE-1131 (Electrical Circuits and Electronics) (2018-2019)

Course: APEE-1131 (Electrical Circuits and Electronics) (2017-2018)

Time: 03 Hours Full Marks: 52.5

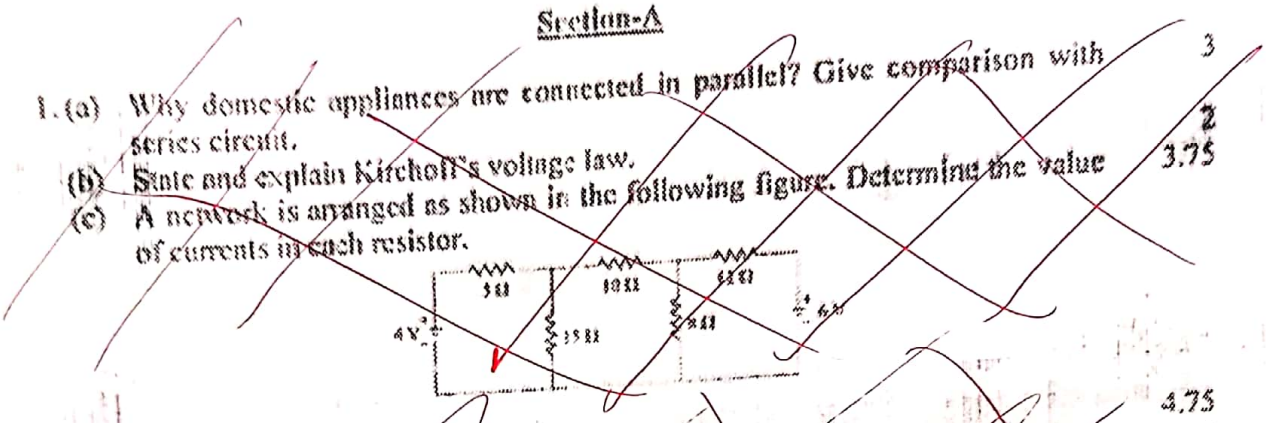
[N.B. Answer 06 (Six) questions taking any 03 (Three) questions from each section. The figures in the right margin indicate full marks.]

Section-A

1. (a) Why domestic appliances are connected in parallel? Give comparison with series circuit. 3

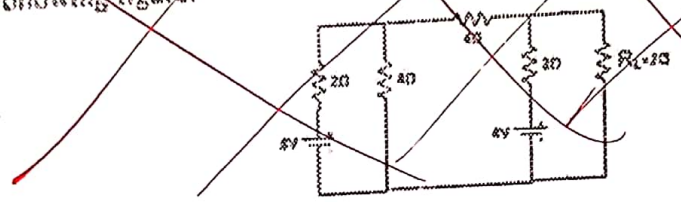
(b) State and explain Kirchoff's voltage law. 2

(c) A network is arranged as shown in the following figure. Determine the value of currents in each resistor. 3.75



2. (a) State and explain superposition theorem. 4.75

(b) Using Thevenin's theorem, find the current through the R_L=2Ω as in the following figure: 4



3. (a) What is a rectifier? Show the circuit diagram of a full wave rectifier. 4

(b) Define voltage regulation (VR) and Peak Inverse Voltage (PIV). 2

(c) A half wave rectifier using Ge diode has secondary emf of 20 V_{p-p}. Diode forward resistance is 0.25 Ohm and load resistance is 1000 Ohm. Find i) Maximum load voltage ii) DC load voltage and iii) efficiency 2.75

4. (a) Why transistor biasing is necessary? Discuss any one method used for transistor biasing. 4

(b) For a certain transistor, I_c= 5.505 mA, I_B=50 μA, I_{co}= 5μA. Determine 2.75

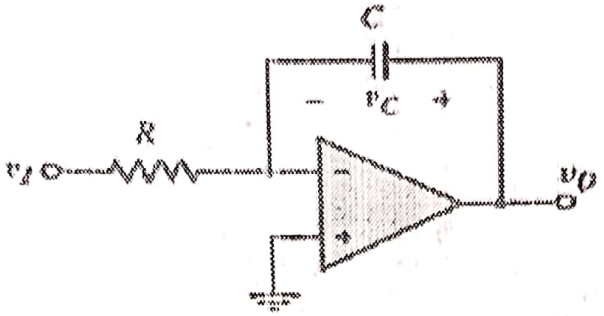
i) The value of α, β and I_E.

ii) The new level of I_B required to make I_c = 10mA.

(c) Show the relationship between α and β for a transistor. 2

Section-B

5. (a) What is an OP-AMP? What are the basic characteristics of an ideal OP-AMP? 3
   (b) What do you mean by virtual ground of an OP-AMP? Explain. 1.75
   (c) A 10mV, 5KHz sinusoidal signal is applied to the input of an OP-AMP integrator as shown below for which R= 50K and C= 2μF. Find the output voltage. 4



6. (a) How is an oscillator different from an amplifier? Mention some of the applications of an oscillator. 4.75
   (b) Design a Hartley oscillator which has to be tunable over 500 kHz to 1000 kHz. The values of the two inductors are 50 μH each. Neglect the effect of mutual inductance. 4

7. (a) What is LCD? Describe the working principle of LCD. 3.75
   (b) What are the advantages of LCD? 2
   (c) What is P-N photodiode? Why it works in reverse biased condition? 3

8. (a) What is a Zener diode? 1
   (b) Explain the V-I characteristics of a Zener diode. 3
   (c) Show that Zener diode can be used as voltage regulator. 4.75

University of Rajshahi
Department of Computer Science of Engineering
B. Sc. (Engg.) Part-I, Odd Semester, Exam - 2018
Course Title: APEE1131 (Electrical Circuit and Electronics)
Total marks: 52.5 Time: 3 Hours

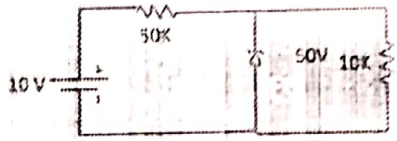
[Answer three questions from each section]

Section-A

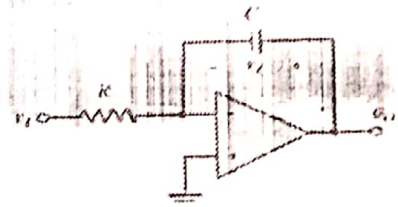
1. a) What is LCD? Describe the working principle of LCD. 3.75
   b) Compare between LED and LCD. 2
   c) What is P-N photodiode? Why it works in reverse biased condition? 3

2. a) Define conductor, semiconductor and insulator in terms of energy band diagram. 3
   b) Discuss the formation process of P-type and N-type extrinsic semiconductor with proper diagram. 3
   c) Consider a specimen of Silicon of length 1.5 cm and area 1 mm2. Calculate the i) conductivity ii) resistivity and iii) resistance of the specimen. (Assume $n_i = 1.5 \times 10^{16}, \mu_e = 0.13\ m^2/v-s$, $\mu_h = 0.05\ m^2/v-s$) 2.75

3. a) What is a PN junction diode? Show the V-I characteristics of a PN junction diode. 3
   b) Differentiate between Zener breakdown and avalanche breakdown. 2.75
   c) For the circuit shown below, find: 3
      i. Output voltage
      ii. Voltage drop across 50 K-Ohm resistor and
      iii. Voltage across the diode.

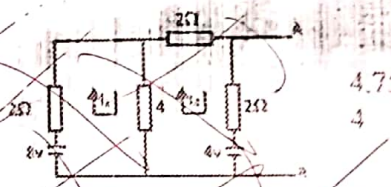


4. a) What is an Op Amp? What are the basic characteristics of an ideal OP-AMP? 3
   b) What do you mean by virtual ground of an op-amp? Explain. 1.75
   c) A 10mV, 5 KHz sinusoidal signal is applied to the input of an OP-AMP integrator as shown below for which R= 100K and C= 1μF. Find the output voltage. 4



Section-B

5. a) State and prove maximum power transfer theorem. 4.75
   b) Use Thevenin's theorem to find the current in a 20Ω load connected between the terminals A and B of the network shown in the figure- 4



6. a) What are the classification of circuit components? Define active components and passive components with examples. 2.75
   b) State and explain Kirchhoff's current law. 3
   c) Write the current division formula when only two resistance is connect in parallel. Two resistor of 4 Ω and 6Ω are connected in parallel. If the total current is 30A, find the individual current through each resistor. 3

7. a) What is an oscillator? What are the conditions for oscillation? 3
   b) What is feedback? 1
   c) Design an Astable Multivibrator whose frequency of oscillation is 5 KHz. Consider $R_1 = R_2$ and $C_1 = C_2$. 4.75

8. a) What is a rectifier? Show the circuit diagram of a full wave rectifier. 4
   b) Define voltage regulation (VR) and Peak Inverse Voltage (PIV). 2
   c) A half wave rectifier using Ge diode has secondary emf of 20 V_{p-p}. Diode forward resistance is 0.25 Ohm and load resistance is 100 Ohm. Find (i) Maximum load voltage (ii) DC load voltage and (iii) efficiency. 2.75

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-I Odd Semester, Examination-2017
Course: APEE 1131 (Electrical Circuit and Electronics)
Time: 3 Hrs. Full Marks: 52.5
[Answer SIX (06) questions taking at least THREE (03) from each Section.]

Section A

1. (a) Write the steps of branch current analysis method — 3
(b) What is passive sign convention? What do you mean by linear and bilateral components? — 2
(c) Find the current through each resistor and voltage drop across each resistor of the circuit using branch current analysis method in figure-1. — 3.75

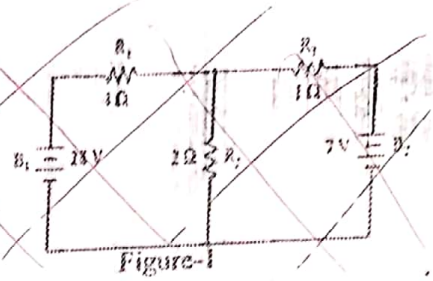


Figure-1

2. (a) What is a PN junction diode? — 1
(b) Define conductor, semiconductor and insulator with energy band diagram. — 1.75
(c) Explain the I-V characteristic of PN junction diode in forward and reverse bias with proper diagram. — 3
(d) How is depletion layer formed in a PN junction diode? Explain with energy band diagram. — 3

3. (a) State and explain Thevenin's theorem. — 4
(b) With reference to the network of figure-2, by applying Thevenin's theorem find: — 4.75
i) The equivalent e.m.f. of the network when viewed from terminals A and B.
ii) The equivalent resistance of the network when looked from terminals A and B.
iii) Current in the load resistance R_L of 15Ω.

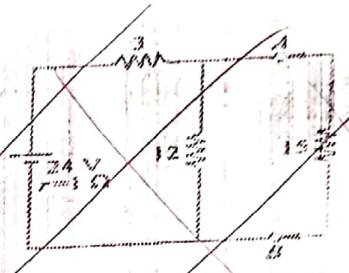


Figure-2

4. (a) What are the different kinds of filters? — 1.75
(b) Deduce an expression for cut-off frequency of a high pass filter. — 3
(c) A filter section is to have a characteristic impedance at zero frequency of 600 Ω and a cut-off frequency at 5 MHz Design (i) a low-pass T section filter, and (ii) a low-pass π section filter to meet these requirements. — 4

Section B

5. (a) Draw the circuit diagram of an astable multivibrator and discuss its operation. — 3
(b) Explain the principle of operation of a photodiode. — 2
(c) Write short notes on LED and LCD. — 3.75

6. (a) What is Bipolar Junction Transistor? Explain the architecture of a Bipolar Junction Transistor. — 4
(b) Determine the value of I_{CQ} and V_{CEQ} for the amplifier shownin figure-3. — 2
(c) What is meant by transistor biasing? Why is it needed? — 2.75

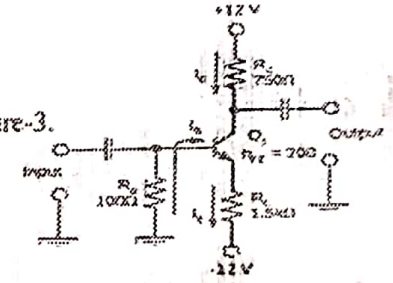


Figure-3

7. (a) Define OP-AMP. What are the characteristics of an ideal OP-AMP? — 2
(b) How can an OP-AMP be used as a differentiator? Explain. — 3.75
(c) What is inverting and non-inverting amplifiers? Explain with necessary figures. — 3

8. (a) What is an oscillator? What are the conditions for oscillation? — 3
(b) Draw the circuit diagram of Hartley oscillator and describe its principle of operation. Derive the equation for oscillation. — 3.75
(c) Explain positive feedback and negative feedback. — 2